

OLIST E-COMMERCE DATA CLEANING AND PREPARATION PROJECT

Executive Summary

Data quality is a critical component of effective analytics and business intelligence. Before meaningful analysis can be conducted, datasets must be validated, standardized, and transformed into a reliable analytical format.

This project focuses on cleaning and preparing the Olist Brazilian E-Commerce Dataset, a multi-table relational database containing customer, order, product, payment, seller, and review information. The objective is to identify and resolve data quality issues, validate dataset integrity, engineer analytical features, and produce clean datasets suitable for future SQL, Power BI, and advanced analytics projects.

1. Business Problem

Organizations depend on high-quality data to support reporting, forecasting, customer analytics, and strategic decision-making.

The Olist e-commerce dataset contains information distributed across multiple related tables. Prior to analysis, several data quality challenges must be addressed, including:

- Missing values
- Duplicate records
- Inconsistent data formats
- Invalid relationships between tables
- Unstructured date fields
- Missing product information

The objective of this project is to transform the raw dataset into a clean, analysis-ready environment while preserving data integrity.

2. Dataset Overview

The project utilizes publicly available Brazilian e-commerce data provided by Olist.

The dataset consists of multiple related tables:

Dataset	Description
Customers	Customer demographic and location information
Orders	Order lifecycle and delivery information
Order Items	Product-level order details
Payments	Payment transactions and methods
Products	Product attributes and category information
Sellers	Seller identification and location data
Reviews	Customer satisfaction and review scores
Category Translation	Product category translations

Together these datasets provide a complete view of customer purchases, payments, products, logistics, and customer satisfaction.

3. Project Objectives

The project was designed to:

- Assess overall data quality
- Validate primary key uniqueness
- Detect duplicate records
- Identify missing values
- Standardize date and text formats
- Validate relationships between tables
- Engineer analytical features
- Produce clean datasets for downstream analytics

4. Data Loading and Inspection

All datasets were imported into Python using Pandas.

Initial inspection included:

- Dataset dimensions
- Column structure review
- Data type verification
- Summary statistics
- Preliminary data quality assessment

This stage established a baseline understanding of the available data.

5. Data Quality Assessment

Several validation checks were performed across all datasets.

Primary Key Validation

Unique identifiers were examined to ensure record uniqueness.

Examples include:

- customer_id
- order_id
- seller_id
- product_id

This process confirmed the integrity of primary key fields.

Duplicate Detection

Duplicate records were investigated across all tables.

Duplicate checks helped ensure that analytical results would not be distorted by repeated observations.

Missing Value Analysis

Missing values were identified and quantified across every dataset.

Special attention was given to:

- Product categories
- Product dimensions
- Delivery timestamps
- Review comments

The impact of missing values was assessed before treatment.

6. Data Cleaning Process

Several cleaning techniques were applied.

Datetime Standardization

Date columns were converted into proper datetime formats.

Examples include:

- Purchase timestamps
- Approval timestamps
- Delivery timestamps
- Review creation dates

This enables reliable time-based analysis.

Missing Value Treatment

Appropriate strategies were used depending on the variable type.

Methods included:

- Logical replacement
- Median imputation
- Category assignment
- Business-rule-based corrections

Product Information Cleaning

Missing product measurements and category values were addressed to improve dataset completeness.

Text Standardization

Text fields were cleaned and standardized to improve consistency.

This included:

- Removing formatting inconsistencies
- Standardizing categories
- Improving readability

7. Relational Data Validation

Since the Olist dataset is composed of multiple related tables, referential integrity checks were performed.

The following relationships were verified:

- Orders linked to customers
- Order items linked to orders
- Payments linked to orders
- Reviews linked to orders
- Products linked to categories

These checks ensured that all records maintained valid relationships across the database.

8. Feature Engineering

Several analytical variables were created to enhance future reporting and business intelligence initiatives.

Delivery Delay Days

Measures the difference between estimated and actual delivery dates.

Total Order Value

Combines product price and freight cost.

Customer Lifetime Value

Estimates customer contribution based on historical purchases.

Purchase Frequency

Measures customer purchasing behaviour.

Average Basket Value

Calculates average spending per order.

Items per Order

Measures purchase volume per transaction.

Revenue per Seller

Aggregates revenue generated by each seller.

Monthly Revenue

Creates a time-based revenue metric for trend analysis.

Monthly Orders

Measures order volume over time.

These engineered features provide a strong foundation for future analytical projects.

9. Final Outputs

The project generated:

- Clean customer dataset
- Clean order dataset
- Clean product dataset
- Clean payment dataset
- Clean seller dataset
- Clean review dataset
- Clean category translation dataset

All outputs were exported as CSV files for future use.

10. Project Deliverables

Deliverables include:

- Jupyter Notebook
- Cleaned datasets
- Documentation
- Data quality validation workflow
- Feature engineering workflow
- Export-ready analytical datasets

11. Conclusion

This project successfully transformed raw Olist e-commerce data into a clean and analysis-ready environment.

Through extensive validation, cleaning, standardization, and feature engineering, the dataset was prepared for future SQL analysis, Power BI dashboard development, customer analytics, and business intelligence reporting.

The resulting datasets provide a reliable foundation for advanced analytics while demonstrating best practices in data preparation and data quality management.